

## Exercise 4 (Joins, UNION, Filtering and Aggregates)

```
1. SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM   employees AS A
INNER JOIN projects AS B ON
A.EmployeeID = B.EmployeeID;
```

| EmployeeID | FirstName | LastName | Department | Salary | ProjectID | ProjectName         | Budget | Status    |
|------------|-----------|----------|------------|--------|-----------|---------------------|--------|-----------|
| 1          | John      | Doe      | IT         | 70000  | 101       | AI development      | 100000 | Completed |
| 1          | John      | Doe      | IT         | 70000  | 103       | Cybersecurity Audit | 75000  | pending   |
| 2          | Alice     | Smith    | HR         | 60000  | 102       | Employee Training   | 50000  | Ongoing   |
| 3          | Bob       | Johnson  | Finance    | 75000  | 104       | Financial Analysis  | 90000  | Ongoing   |
| 5          | Emma      | Wilson   | Sales      | 65000  | 105       | Market Expansion    | 65000  | Completed |
| 6          | Michael   | Clark    | Finance    | 80000  | 106       | Risk Management     | 80000  | pending   |

```
2. SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
```

FROM employees AS A  
LEFT JOIN Projects AS B  
ON A.employeeID = B.employeeID;

| EmployeeID | Firstname | LastName | Department | Salary | ProjectID | ProjectName          | Budget | Status    |
|------------|-----------|----------|------------|--------|-----------|----------------------|--------|-----------|
| 1          | John      | Doe      | IT         | 70000  | 101       | AI Development       | 100000 | Completed |
| 1          | John      | Doe      | IT         | 70000  | 103       | Cyber Security Audit | 75000  | Pending   |
| 2          | Alice     | Smith    | HR         | 60000  | 102       | Employee Training    | 50000  | Ongoing   |
| 3          | Bob       | Johnson  | Finance    | 75000  | 104       | Financial Analysis   | 90000  | Ongoing   |
| 4          | David     | Brown    | IT         | 72000  | NULL      | NULL                 | NULL   | NULL      |
| 5          | Emma      | Wilson   | Sales      | 65000  | 105       | Market Expansion     | 65000  | Completed |
| 6          | Michael   | Clark    | Finance    | 80000  | 106       | Risk Management      | 80000  | Pending   |

3. SELECT ProjectID,  
ProjectName,  
Budget,  
Status,  
EmployeeID,  
Firstname,  
LastName,  
Department,  
Salary

FROM Employees AS A  
RIGHT JOIN Projects AS B  
ON A.Employees ID = B.Employees ID;

| ProjectID | ProjectName         | Budget | Status    | EmployeeID | Firstname | LastName | Department | Salary |
|-----------|---------------------|--------|-----------|------------|-----------|----------|------------|--------|
| 101       | AI Development      | 100000 | Completed | 1          | John      | Doe      | IT         | 70000  |
| 102       | Employee Training   | 50000  | Ongoing   | 1          | John      | Doe      | IT         | 70000  |
| 103       | Cybersecurity Audit | 75000  | Pending   | 2          | Alice     | Smith    | HR         | 60000  |
| 104       | Financial Analysis  | 90000  | Ongoing   | 3          | Bob       | Johnson  | Finance    | 75000  |
| 105       | Market Expansion    | 65000  | Completed | 5          | Emma      | Wilson   | Sales      | 65000  |
| 106       | Risk Management     | 80000  | Pending   | 6          | Michael   | Clark    | Finance    | 80000  |

4. SELECT EmployeeID,  
 FirstName,  
 LastName,  
 Department,  
 Salary,  
 ProjectID,  
 ProjectName,  
 Budget,  
 Status

FROM Employees AS A

FULL OUTER JOIN projects AS B

ON A.EmployeeID = B.EmployeeID;

| EmployeeID | FirstName | LastName | Department | Salary | ProjectID | ProjectName          | Budget | Status    |
|------------|-----------|----------|------------|--------|-----------|----------------------|--------|-----------|
| 1          | John      | Doe      | IT         | 70000  | 101       | AI Development       | 100000 | Completed |
| 1          | John      | Doe      | IT         | 70000  | 103       | Cyber Security Audit | 75000  | Pending   |
| 2          | Alice     | Smith    | HR         | 60000  | 102       | Employee Training    | 50000  | Ongoing   |
| 3          | Bob       | Johnson  | Finance    | 75000  | 104       | Finance Analysis     | 90000  | Ongoing   |
| 4          | David     | Brown    | IT         | 72000  | NULL      | NULL                 | NULL   | NULL      |
| 5          | Emma      | Wilson   | Sales      | 65000  | 105       | Market Expansion     | 65000  | Completed |
| 6          | Michael   | Clark    | Finance    | 80000  | 106       | Risk Management      | 80000  | Pending   |

7. SELECT EmployeeID,  
 FirstName,  
 LastName,  
 Department,  
 Salary

FROM Employees

WHERE salary > 70000;

| EmployeeID | FirstName | lastName | Department | Salary |
|------------|-----------|----------|------------|--------|
| 3          | Bob       | Johnson  | Finance    | 75000  |
| 4          | David     | Brown    | IT         | 72000  |
| 6          | Michael   | Clark    | Finance    | 80000  |

```

8. SELECT EmployeeID,
      FirstName,
      LastName,
      Department,
      Salary
FROM   Employees
WHERE  Department IN ('IT', 'Finance');

```

| EmployeeID | FirstName | lastName | Department | Salary |
|------------|-----------|----------|------------|--------|
| John 1     | John      | Doe      | IT         | 70000  |
| Bob 3      | Bob       | Johnson  | Finance    | 75000  |
| David 4    | David     | Brown    | IT         | 72000  |
| Michael 6  | Michael   | Clark    | Finance    | 80000  |

```

9. SELECT projectID,
      projectName,
      Budget,
      status
FROM   projects
WHERE  status != 'completed';

```

| ProjectID | ProjectName         | Budget | status  |
|-----------|---------------------|--------|---------|
| 102       | Employee Training   | 50000  | Ongoing |
| 103       | Cybersecurity Audit | 75000  | Pending |
| 104       | Financial Analysis  | 90000  | Ongoing |
| 106       | Risk Management     | 80000  | pending |

```

10. SELECT Project ID,
           ProjectName,
           Budget,
           Status
FROM   projects
WHERE  Budget > 70000
AND    Status != 'completed';

```

| Project ID | ProjectName         | Budget | Status  |
|------------|---------------------|--------|---------|
| 103        | Cybersecurity Audit | 75000  | Pending |
| 104        | Financial Analysis  | 90000  | Ongoing |
| 106        | Risk Management     | 80000  | Pending |

```

11. SELECT EmployeeID,
           FirstName,
           LastName,
           Department,
           Salary,
           City
FROM   Employees WHERE City IN ('New York', 'Toronto') ORDER BY Salary DESC;

```

| EmployeeID | FirstName | LastName | Department | Salary |
|------------|-----------|----------|------------|--------|
| 6          | Michael   | Clark    | Finance    | 80000  |
| 3          | Bob       | Johnson  | Finance    | 75000  |
| 1          | John      | Doe      | IT         | 70000  |

```

12. SELECT EmployeeID,
           FirstName,
           LastName,
           Department,
           Salary
FROM   Employees

```

ORDER BY Salary DESC  
LIMIT 3;

| EmployeeID | FirstName | LastName | Department | Salary |
|------------|-----------|----------|------------|--------|
| 6          | Michael   | Clark    | Finance    | 80000  |
| 3          | Bob       | Johnson  | Finance    | 75000  |
| 4          | David     | Brown    | IT         | 72000  |

13. SELECT Department,  
SUM(salary) AS TotalsSalary  
FROM Employees  
GROUP BY Department  
ORDER BY TotalsSalary DESC;

| Department | TotalsSalary |
|------------|--------------|
| Finance    | 155000       |
| IT         | 142000       |
| Sales      | 65000        |
| HR         | 60000        |

14. SELECT City,  
AVG(salary) AS AverageSalary  
FROM Employee  
GROUP BY City  
HAVING AVG(salary) > 65000;

| City     | AverageSalary |
|----------|---------------|
| New York | 75000         |
| Toronto  | 75000         |
| London   | 72000         |

15. SELECT Department,  
COUNT(EmployeeID) AS EmployeeCount  
FROM Employees  
GROUP BY Department  
HAVING COUNT(EmployeeID) > 1;

| Department | EmployeeCount |
|------------|---------------|
| IT         | 2             |
| Finance    | 2             |

16. SELECT status  
COUNT (ProjectID) AS projectCount  
FROM Projects  
GROUP BY status  
HAVING COUNT(projectID) ≥ 2;

| Status    | ProjectCount |
|-----------|--------------|
| Completed | 2            |
| Ongoing   | 2            |
| Pending   | 2            |

17. SELECT EmployeeID,  
FirstName,  
LastName,  
SUM(Budget) AS TotalProjectBudget  
FROM Employees AS A  
JOIN Projects AS B ON A.EmployeeID = B.EmployeeID  
GROUP BY EmployeeID, FirstName, LastName  
HAVING SUM(Budget) > 150000;

| EmployeeID | FirstName | LastName | TotalProjectBudget |
|------------|-----------|----------|--------------------|
| 1          | John      | Doe      | 175000             |